ESSENTIAL EXTENSIONS CHARACTERISED BY NON-ANNIHILATED MULTIPLES

THE STACKS PROJECT AUTHORS

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Lemma 0.1. *Let R be a ring. Let $M \subset N$ be R-modules. The following are equivalent*

(1) *$M \subset N$ is an essential extension,*
(2) *for all $x \in N$ nonzero there exists an $f \in R$ such that $fx \in M$ and $fx \neq 0$.*

Proof. Assume (1) and let $x \in N$ be a nonzero element. By (1) we have $Rx \cap M \neq 0$. This implies (2).

Assume (2). Let $N' \subset N$ be a nonzero submodule. Pick $x \in N'$ nonzero. By (2) we can find $f \in R$ with $fx \in M$ and $fx \neq 0$. Thus $N' \cap M \neq 0$. □

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